Exercise 02 - SQL Foundations
SQL Aggregate Functions and Operators

The students Table

01 SELECT DISTINCT department
FROM students;

department
IT
HR
Finance

02 SELECT department, ~~AVG~~
AVG (age) AS avg-age
FROM students
GROUP BY department;

department	avg-age
IT	20,5
HR	22
Finance	23

03 SELECT department
COUNT (*) AS student-count
FROM students
GROUP BY department
HAVING COUNT (*) > 1;

department	student_count
IT	2
HR	2

Q4
```
SELECT student_id, name, age, department
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q5
```
SELECT student_id, name, age, department
FROM students
WHERE department = 'IT' OR 'HR' AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Q6 The course Table
```
SELECT SUM (credits) AS total_credits
FROM course
GROUP BY gender
HAVING total_credits > 5
```

department	total_credits
IT	11

(07) SELECT course_id, course_name, department, credits
FROM courses , WHERE credits < 4
GROUP BY course_name;

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

(08) SELECT course id, course_name, credits
FROM courses
ORDER BY DESC
LIMIT 3;

course_id	course_name	credits
102	Python	4
103	Data Science	4
103	SQL Basics	3

(09) SELECT MAX (grade) AS Max_grade
FROM enrollments;

Max_grade
90

SELECT MIN (grade) AS min_grade
FROM enrollments;

Min_grade
78

```
SELECT AVG (grade) AS avg_grade
FROM enrollments
```

avg_grade
84,6

(10)
```
SELECT course id, COUNT (*) AS enrollment count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

(11)
```
SELECT department
       SUM (salary) AS total_salary
       SUM (bonus) AS total_bonus
FROM salaries ;
```

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

(12)
```
SELECT department
       AVG (salary) AS avg_salary
FROM salaries
WHERE salary > 55000
```

```
GROUP BY department
HAVING avg_salary > 55000;
```

department	avg_salary
IT	61000
Finance	70000

(13)
```
SELECT employee_id, name, salary, bonus
       (salary + bonus) AS total compensation
FROM salaries
WHERE (salary + bonus) > 60000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	50000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

The projects Table

(14)
```
SELECT department
       SUM (total_budget) AS total budget
       AVG (avg_budget) AS avg_budget
FROM projects
GROUP BY department
HAVING avg_budget > 70000
```

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

(15) SELECT project_id, product_name, department, budget
FROM salaries
WHERE budget BETWEEN 50 000 AND 70000
AND department <> 'Marketing';

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000